

# Machine Learning

25SC2107E

|               |         |         |                     |                |     |
|---------------|---------|---------|---------------------|----------------|-----|
| L-T-P-S       | 2-0-2-4 | Credits | 4                   | Contact Hrs/Wk | 8   |
| Trimester     | T04     | Mode    | PBL (Project-Based) | Category       | ESC |
| Pre-requisite | CFAI    |         |                     |                |     |

## Course Outcomes (COs)

|            | Statement                                                                                                                                                                                                                                                                    | BTL            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>CO1</b> | Analyze the end-to-end ML system lifecycle — data → features → training → evaluation → deployment → monitoring — and trace one prediction request from a deployed model back through every preceding lifecycle stage.                                                        | BTL4 · Analyze |
| <b>CO2</b> | Apply linear supervised-learning models — linear regression, ridge, lasso, elastic net, logistic regression, multinomial logistic regression — and reason about regularisation, the bias-variance trade-off, and feature scaling.                                            | BTL3 · Apply   |
| <b>CO3</b> | Apply tree-based supervised-learning models — decision trees, random forests, gradient-boosted trees (XGBoost, LightGBM) — and reason about feature importance, splitting criteria, and ensemble variance reduction.                                                         | BTL3 · Apply   |
| <b>CO4</b> | Apply unsupervised-learning techniques — k-means and hierarchical clustering, density-based clustering (DBSCAN), dimensionality reduction (PCA, t-SNE, UMAP), and anomaly detection — to discover structure in unlabelled data.                                              | BTL3 · Apply   |
| <b>CO5</b> | Analyze model performance using cross-validation, train/validation/test splits, classification metrics (accuracy, precision, recall, F1, ROC-AUC, PR-AUC), regression metrics (RMSE, MAE, R <sup>2</sup> ), calibration, and hyperparameter search (grid, random, Bayesian). | BTL4 · Analyze |
| <b>CO6</b> | Apply ML-engineering practices — feature stores, training-serving skew avoidance, model packaging, REST-based serving, basic monitoring for drift and performance degradation — to deploy a production-style ML system.                                                      | BTL3 · Apply   |

## CO–PO / PSO Articulation Matrix

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PSO1 |
|------------|-----|-----|-----|-----|-----|------|
| <b>CO1</b> | H   | H   | H   | –   | –   | H    |
| <b>CO2</b> | H   | H   | H   | –   | –   | H    |
| <b>CO3</b> | H   | –   | H   | –   | –   | H    |
| <b>CO4</b> | H   | M   | –   | –   | –   | H    |
| <b>CO5</b> | H   | H   | –   | H   | –   | H    |
| <b>CO6</b> | H   | –   | H   | –   | H   | H    |

H = High (3) · M = Moderate (2) · L = Low (1) · blank = no significant correlation

## Module-wise Syllabus

|           | Title & Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CO  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>M1</b> | <b>PlacementPredict as a System — The ML Lifecycle</b><br>A working prediction service running on day one — students issue requests against PlacementPredict and see predictions before any internals.; The ML system lifecycle: raw data → cleaned dataset → engineered features → training set → trained model → packaged artifact → served endpoint → monitored production → retraining loop.; Where each lifecycle stage actually lives in code: data pipeline scripts, feature engineering modules, training notebooks, model registry, serving framework, monitoring dashboard.; The supervised vs unsupervised vs reinforcement learning split; what PlacementPredict is (supervised classification) and what it is not.; Tracing one prediction end-to-end: HTTP request → feature lookup → model inference → response; what changes for a batch prediction.; The training-serving boundary: the split that causes the most production bugs (training-serving skew); why feature stores exist.; A reading of the PlacementPredict codebase: data ingestion, feature pipeline, training script, serving FastAPI app, monitoring hooks.; What this course owns vs what comes downstream: classical ML and ML engineering here; deep learning in DLAC; generative models in GALE; LLMs in NLPL. | CO1 |
| <b>M2</b> | <b>Supervised Learning — Linear Models at Depth</b><br>Linear regression revisited from CFAI: closed-form normal equations, gradient-descent training, the comparison.; Regularisation: ridge (L2), lasso (L1), elastic net; the geometric picture of why L1 induces sparsity.; Logistic regression for binary classification: the sigmoid, the cross-entropy loss, the decision boundary as a hyperplane.; Multinomial logistic regression (softmax regression) for multi-class problems.; Feature scaling and standardisation: when it matters (gradient descent), when it doesn't (tree models in M3); the StandardScaler and MinMaxScaler.; Categorical features: one-hot encoding, ordinal encoding, target encoding, embedding-based encoding at preview level.; Handling missing values: deletion, mean/median imputation, model-based imputation, MissingIndicator features.; A worked example end-to-end: predict student placement outcome from academic features using ridge logistic regression; train, validate, interpret coefficients.                                                                                                                                                                                                                                                | CO2 |
| <b>M3</b> | <b>Supervised Learning — Tree-Based Models</b><br>Decision trees: greedy top-down construction, splitting criteria (Gini impurity, entropy, MSE for regression), stopping rules.; Tree pruning: pre-pruning by depth/leaf-size, post-pruning by cost-complexity (CCP); the bias-variance angle.; Random forests: bagging, feature subsampling, out-of-bag error, the variance-reduction argument.; Boosting: AdaBoost at intuition level, gradient boosting as functional gradient descent.; XGBoost and LightGBM as the modern industrial workhorses: histogram-based splitting, leaf-wise growth, regularisation.; Feature importance: gini importance, permutation importance, SHAP values at preview level.; Tree models on tabular data: why they routinely beat neural networks on structured datasets (the Kaggle reality).; A worked example end-to-end: same placement-prediction task with a gradient-boosted tree; compare against linear baseline; interpret feature importance.                                                                                                                                                                                                                                                                                                         | CO3 |
| <b>M4</b> | <b>Unsupervised Learning</b><br>K-means revisited from CFAI with the modern Lloyd-algorithm view; choosing K via the elbow method and the silhouette score.; K-means++ initialisation revisited; mini-batch k-means for large datasets.; Hierarchical clustering: agglomerative bottom-up, the linkage choices (single, complete, average, Ward), reading dendrograms.; Density-based clustering with DBSCAN: $\epsilon$ and minPts, identifying outliers naturally, why DBSCAN handles non-convex clusters.; Dimensionality reduction revisited: PCA from CFAI in production-style usage; pitfalls when applied without standardisation.; t-SNE and UMAP as the visualisation workhorses; their hyperparameters; why distances in t-SNE/UMAP space are not meaningful.; Anomaly detection: isolation forest, one-class SVM at intuition level, autoencoder-based anomaly detection at preview level (deep dive in DLAC).; Practical pipeline: when to cluster vs when to use a supervised model; cluster labels as features for a downstream supervised model.                                                                                                                                                                                                                                      | CO4 |

|           | Title & Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CO  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>M5</b> | <b>Model Evaluation, Selection, and Calibration</b><br>Train/validation/test split discipline: the three-way split, why a test set must be held out from every modelling decision.; k-fold and stratified k-fold cross-validation; nested CV when both hyperparameter selection and unbiased evaluation are needed.; Classification metrics: accuracy (and when it lies), precision/recall/F1, the confusion matrix, ROC-AUC, PR-AUC; class-imbalance impact.; Regression metrics: RMSE, MAE, MAPE, $R^2$ ; the choice between RMSE and MAE depending on outlier sensitivity.; Probability calibration: why a 0.7 prediction should mean "70% likely"; Platt scaling, isotonic regression; reliability diagrams.; Hyperparameter search: grid search, random search (Bergstra-Bengio), Bayesian optimisation with Gaussian processes at preview level.; Learning curves and validation curves: diagnosing under-fitting, over-fitting, data scarcity from a single plot.; Statistical significance for model comparison at intro level: McNemar's test for classifiers, paired bootstrap for regressors. | CO5 |
| <b>M6</b> | <b>ML Engineering — From Notebook to Production</b><br>The notebook-to-production gap: why a Jupyter notebook is not a deployable system; the ML-engineering rituals that close the gap.; Feature engineering at scale: feature stores (Feast at conceptual level), the offline-online split, point-in-time correctness.; Training-serving skew: the canonical bug class in production ML; how to detect and prevent it.; Model packaging: pickle pitfalls, ONNX as the portable format, container-based serving.; Serving frameworks at intro level: FastAPI for REST, gRPC for low-latency, batch vs online prediction.; Monitoring in production: predictive performance, latency, throughput, data drift, concept drift, label drift.; Retraining strategies: scheduled retraining, drift-triggered retraining, online learning at preview level.; A glance at MLOps platforms: MLflow for experiment tracking, DVC for data versioning, the basic engineer's toolkit. (Deep MLOps and platform engineering belong to GALE and SDSR.)                                                                | CO6 |

## Reference Books

| # | Title                                                                                              | Authors                                                                          | Publisher          | Year                                    |
|---|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------|-----------------------------------------|
| 1 | <b>Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow</b>                          | Aurélien Géron                                                                   | O'Reilly           | 2022 (3rd Ed.)                          |
| 2 | <b>The Elements of Statistical Learning — Data Mining, Inference, and Prediction</b>               | Trevor Hastie, Robert Tibshirani & Jerome Friedman                               | Springer           | 2017 (2nd Ed., corrected 12th printing) |
| 3 | <b>An Introduction to Statistical Learning — with Applications in Python</b>                       | Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani & Jonathan Taylor | Springer           | 2023                                    |
| 4 | <b>Pattern Recognition and Machine Learning</b>                                                    | Christopher M. Bishop                                                            | Springer           | 2006                                    |
| 5 | <b>Designing Machine Learning Systems — An Iterative Process for Production-Ready Applications</b> | Chip Huyen                                                                       | O'Reilly           | 2022                                    |
| 6 | <b>Building Machine Learning Powered Applications</b>                                              | Emmanuel Ameisen                                                                 | O'Reilly           | 2020                                    |
| 7 | <b>Machine Learning Engineering</b>                                                                | Andriy Burkov                                                                    | True Positive Inc. | 2020                                    |